Instance-Space Analysis for the Heterogeneous On-Demand Bus Routing Problem

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Abstract

This report records the design, implementation, and preliminary evidence for an instance-space analysis (ISA) of the static heterogeneous on-demand bus routing problem (H-ODBRP). The study uses simulated annealing (SA) and a greedy construction (GR) as the current comparison pair. A pilot of 24 generated instances paired with three fleets produced 72 solver cases. SA was better on 70 of 72 cases; the two-level pilot still has a gap along the time-window dimension. A seed defect was identified and fixed, and the reported count of distinct mean OD distances rose from 6 to 24. A rewritten exact model and a checker replace an original model that disagreed with the heuristic rules in 9 of 11 scenarios. Full-set generation is not yet established: the station matrix job, memory sizing, a third algorithm, and several design choices remain [**OPEN**]. The proposed full design uses a Latin hypercube over the Capital Region of Denmark, excluding Bornholm, and grows in batches as the instance space expands.

1 Summary

The study asks how instance characteristics relate to the performance of algorithms for the static H-ODBRP. Instances are described by demand, time window, fleet, and network features, then projected into a low-dimensional instance space. SA and GR form the present performance matrix. The exact Gurobi model is reserved for validation on very small cases; a MIP-based large-neighbourhood matheuristic is intended to show where SA can improve, not to act as a third ISA algorithm.

Work completed includes a 72-case Copenhagen pilot, a correction to request seeding, a network cut-out for Bornholm, a nearest-node cache, solver input fixes, an exact-model checker, a rewritten exact model, and HPC job scripts. The pilot suggests that SA is usually better than GR: SA wins on 70 of 72 cases. GR is competitive in a narrow corner: tight time windows and the smallest pilot fleet. The pilot projection retains a gap associated with the two time-window settings. The available feature CSVs do not independently reproduce the documented pre-fix count of six distinct mean OD distances; the post-fix CSV does contain 24.

The proposed full design covers 100–1000 requests in a two-hour period and draws instance settings and fleets using Latin hypercubes. The region, period, and ranges remain [**PROPOSED**]; the author has the final decision. The new exact model agrees with the heuristic feasibility checks in all 11 tested scenarios, while the original model disagreed in 9. On rerun, the new model proved the five-request case in 2.5 seconds, but had a 25% gap at the limit for 10 requests and did not prove the 15-request case.

The full instance set is still blocked by unresolved generation cost and memory questions, an unrun station-matrix build, and the missing shared-matrix feature step. Other open work includes the third ISA algorithm, the matheuristic HPC test, solver resource sizing, and final design choices. These results are preliminary, and the one-sided SA–GR result must be tested on the wide design.

2 Study design

2.1 Network, period, and instance settings

The study network is the whole Capital Region of Denmark without Bornholm. Instances use a two-hour planning period beginning between 12:00 and 19:30. The current design draws 50–500 requests per hour (100–1000 in each instance). All instance ranges below are the present values in `isa_set/design.json`; the settings document describes them as **[PROPOSED]**unless stated otherwise.

Table 1: Study settings, comparison with the earlier papers, and current status. [H] and [P] refer to the two papers in [1, 2].

Setting	Study range	Paper range	Status
Network	Capital Region, excluding Bornholm	Chicago [H]; city scenarios [P]	**[PROPOSED]**
Planning period	2 hours	1 hour [H]; 2–7 hours [P]	**[PROPOSED]**
Start time	12:00–19:30	Commuting and event windows [P]	**[PROPOSED]**
Requests	50–500 per hour (100–1000 per instance)	300–3000 per hour [H]; 5000–20000 per scenario [P]	**[PROPOSED]**; **[OPEN-S1]**
Peak spread	10–60 min standard deviation	About 15 min plus 40% uniform in one [P] scenario	**[PROPOSED]**
Urgency mean	5–40 min	5, 10, 15, 30 min [H]	**[PROPOSED]**
Urgency standard deviation	1–10 min	0, 5, 10 min [H]	**[PROPOSED]**
Arrival slack	5–60 min	Not varied	**[PROPOSED]**
Maximum walking time	550–600 s	Generator default; 550–600 s in earlier settings	**[PROPOSED]**; **[OPEN-S2]**
Walking speed	1.38889 m/s	Same	**[PROPOSED]**
Direct trip time	300–3600 s	Same lower bound; study upper bound raised for suburbs	**[PROPOSED]**
Vehicle speed	20 km/h	Same	**[PROPOSED]**
Fleet size	5–40% of requests	10–21% [H]	**[PROPOSED]**
Eight-seat share	0–100%; remainder four-seat	Four- and eight-seat mix performed well [H]	**[PROPOSED]**
Fleet draws	3 per generated instance	Fleet fixed per instance [P]	**[PROPOSED]**
Depot	One station unused by requests, selected after generation	Several depots [H]	**[DECIDED]**; **[OPEN-S3]**

The network build produced 3515 stations and no station on Bornholm. The station-to-station travel-time matrix is intended to be computed once for the network, in a separate HPC job. The number of batches is not fixed: batches of 100 are proposed until fewer than about 5% of new instances are far from existing ones, or a cap is reached. The cap remains **[OPEN-S4]**.

2.2 Axes of the instance space

The feature list confirms 18 features for the static ISA. The last three features in the list are dynamic-only; the two network features at the end are not implemented for this study.

Table 2: Static ISA features and their short definitions.

Feature	Definition
`n_requests`	Number of requests.
`demand_density`	Requests per square kilometre of service area.
`spatial_cluster_coeff`	Clark–Evans nearest-neighbour ratio for request origins.
`od_distance_mean`	Mean origin–destination driving distance.
`od_distance_std`	Standard deviation of origin–destination driving distance.
`tw_width_mean`	Mean time-window width.
`tw_width_std`	Standard deviation of time-window width.
`tw_overlap_ratio`	Fraction of request pairs with overlapping time windows.
`tight_tw_fraction`	Fraction of requests with window width below 20 minutes.
`tw_tightness_factor`	Mean time-window width divided by minimum trip time.
`fleet_size`	Number of vehicles relative to request count.
`capacity_cv`	Standard deviation of vehicle capacities divided by their mean.
`fleet_utilization_ratio`	Total fleet capacity divided by request count.
`request_vehicle_ratio`	Request count divided by fleet size.
`station_density`	Bus stops per square kilometre.
`bsa_flexibility`	Mean available pick-up or drop-off stops per passenger.
`avg_station_degree`	Mean degree of bus stops in the road graph.
`coverage_ratio`	Fraction of the service area within 500 m of a bus stop.

2.3 Fleets, algorithms, and performance

Each generated request instance is paired with three fleets. Fleet size is drawn from 5–40% of requests; the eight-seat share is drawn from 0–100%, with the remainder in four-seat vehicles. The depot is selected after generation at a station not used by a request.

The objective is total passenger ride time plus four times direct travel time for each request sent to a third party. Operator cost is not in the objective; vehicle use and driving time may be recorded as post-hoc measures.

The performance matrix currently compares SA with GR. The exact Gurobi model is used only for validation on 5–15 requests. A third ISA algorithm is **[OPEN]**: the options are another SA configuration or a max-ALNS variant. The matheuristic is not a candidate ISA algorithm.

3 What was implemented

3.1 Generator changes

Table 3: ISA-specific generator changes against the original REQreate.

Change	File(s)	Why	Effect
Config-dependent request seeds	`REQreate/passenger_requests.py`	All configs previously reused locations.	24 distinct `od_distance_mean` values after the fix; pre-fix count is not reproduced by available CSVs.

Change	File(s)	Why	Effect
Place exclusions	`retrieve_network.py`, `input_json.py`, `schema/config.schema.json`, `cli.py`	Exclude the island of Bornholm from the region polygon.	The network build reported no Bornholm stations.
Polygon-scoped points of interest	`osm_features.py`, station, school, and hospital retrieval	Place-name lookup had allowed an outside station to leak in.	3515 stations; none on Bornholm.
Cached nearest-node index	`REQreate/nearest.py`, `passenger_requests.py`, `network_class.py`	Nearest-node calls repeatedly converted the full graph.	Unit test matches OSMnx on 30 random points. Profile timings improved after the cache; memory measurement is suspect.
ISA feature export	`REQreate/isa_features_hook.py`	Record one feature row per generated instance.	Feature errors are logged without stopping generation.
Station-matrix chunked searches	`REQreate/compute_distance_matrix.py`, `isa_set/station_matrix.py`	Avoid a full graph row cache and Python all-pairs loop.	Same whole-second values in a unit test; separate HPC run remains [**OPEN**].
Pilot v2 sampling	`pilot/run_pilot.py`	Use a real 7:30 peak, replicas, and three fleet sizes.	Replaces the earlier flattened demand peak.

3.2 Solver changes

- The `--greedy_only` option runs GR from the SA binary, avoiding a separate code base.
- The solver now reads travel-time matrices as directed (row = from, column = to); it no longer symmetrises them.
- The type-3 request reader converts walking distance in metres to seconds using the walking speed.
- The station limit was raised from 2000 to 4000 for the 3515-station region. This ODBRP_P change is uncommitted; the author must commit it [**OPEN-A4**].
- The original IP model was replaced by `model/MDHODBRP_model_v2.cpp`. Its checker, `model/check_model.py`, tests hand-made routes without a Gurobi dependency.
- A MIP-based LNS matheuristic re-solves parts of SA solutions using v2. Its HPC run is [**OPEN-A2**].

3.3 HPC tooling

The set tooling includes `isa_set/design.json` for ranges, `isa_set/make_batch.py` for Latin-hypercube batches, single-core-per-config generation scripts, and a profiling script. Pilot solver job generation and result collection are provided by `pilot/make_solver_jobs.py` and `pilot/collect_results.py`. The station-matrix implementation is written, but its HPC run and measured resource profile remain [**OPEN-G3**].

4 Tests and preliminary results

4.1 Pilot instance space and SA versus GR

The pilot contains 12 configurations with two replicas (24 request instances), each paired with three fleets (72 solver cases). It uses 200, 500, or 1000 requests; peak standard deviations of 15 or 30 minutes; tight or loose time-window settings; and fleets at 10%, 20%, or 30% of request count. The instance-space plot shows the two time-window groups and colours cases by the SA-to-GR ratio. The documented gap is tied to having only two levels on this axis; the full design uses continuous ranges [**PROPOSED**].

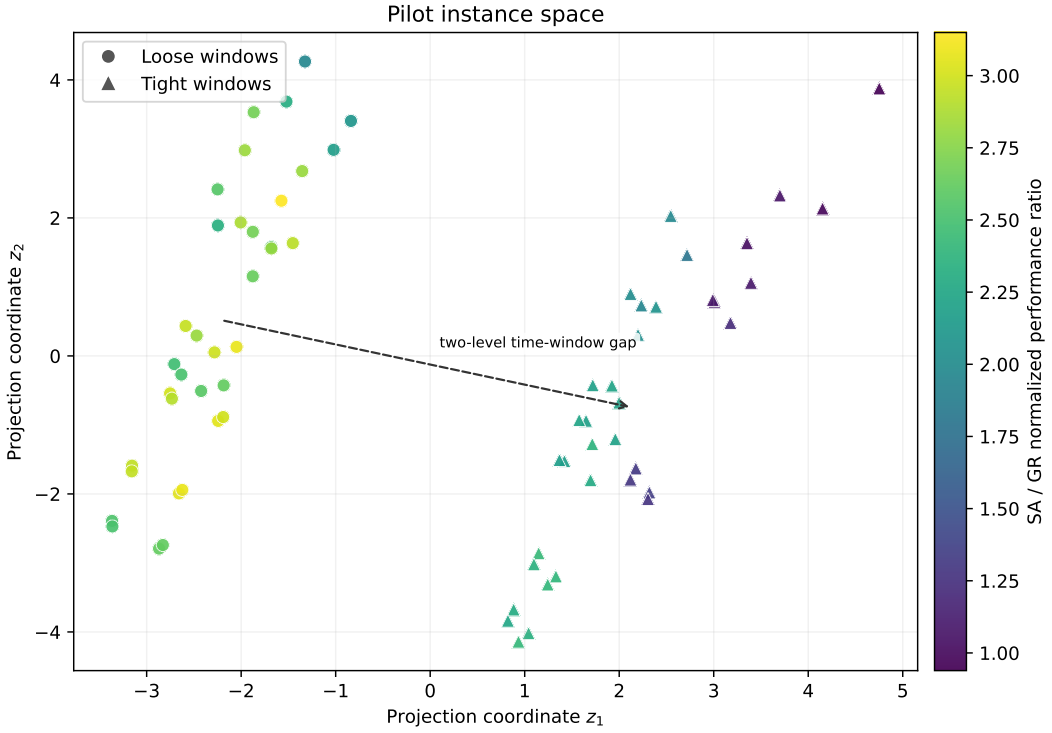


Figure 1: Pilot projection `z1`–`z2`. Colour is the SA/GR ratio of the normalized performance values (above 1 favours SA); marker shape separates tight and loose time windows. The arrow marks their centroid separation and the two-level time-window gap. Data: `pilot/work/results/instance_space.csv` and `pilot/work/results/perf_matrix.csv`.

Across all cases, SA is better on 70 of 72. GR is competitive only for tight windows with the smallest fleet (8 of 72). This one-sided result may make the pilot's algorithm-selection story thin. Whether it persists in the wide design is [**OPEN-A6**].

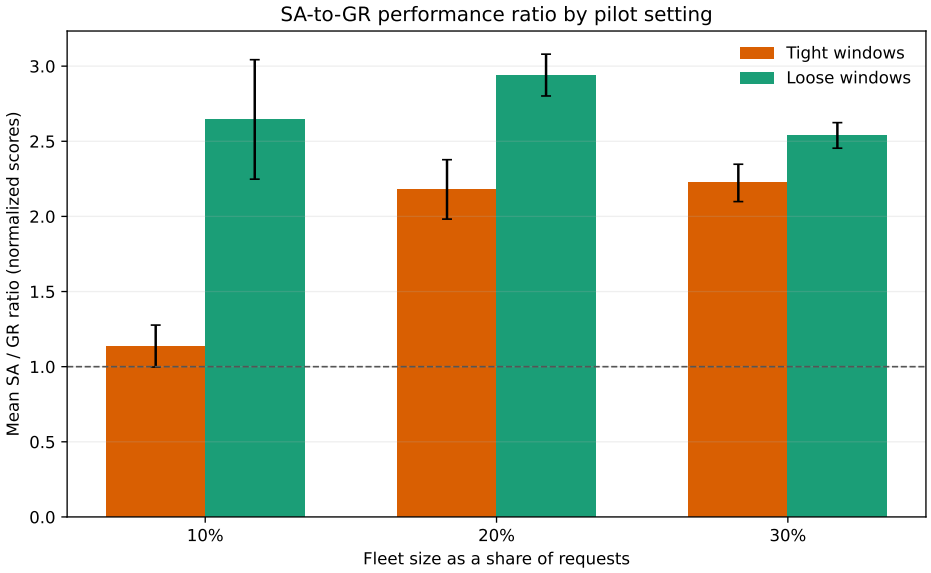


Figure 2: Mean SA/GR normalized performance ratio by pilot time-window setting and fleet size; error bars show one standard deviation across 12 cases in each group. A ratio above 1 favours SA. Data: `pilot/work/results/perf_matrix.csv`.

The seed issue is documented as 6 distinct `od_distance_mean` values before the fix and 24 after it. The available pre-fix-labelled file has only 24 rows and 3 distinct values; the other 72-row files have 24. The pre-fix value of 6 cannot be reproduced from these files, so no before/after bar chart is shown. The post-fix value of 24 is present in `pilot/work/results/features.csv`.

4.2 Model checker and exact model

The checker compared the original and rewritten models with the heuristic feasibility rules in 11 scenarios. The original model mismatched in 9; v2 mismatched in none. The documented v1 defects were construction infeasibility, capacity effectively `C-2`, shifted travel-time columns, no third-party option, and an objective inconsistent with the heuristic.

Table 4: Exact model v2 on small instances. First-run values are from the available raw record before valid ride lower bounds and other improvements. Rerun figures are reported by session in `docs/02_tests_and_results.md`; the rerun objective and bound were not tabulated there.

Requests	Run	Objective	Bound	Gap	Time (s)	Result
5	First	7284.99	170.376	97.66%	3600.02	Time limit
5	Rerun	Not tabulated	Not tabulated	0% (optimal)	2.5	Optimal after valid ride lower bounds
10	First	16261	0	100%	3600.07	Time limit
10	Rerun	Not tabulated	Not tabulated	25%	3600 limit	Gap reported at limit
15	First	39034	0	100%	3600.35	Time limit
15	Rerun	Not tabulated	Not tabulated	Not reported	3600 limit	Not solved to optimality

Table note: Rerun status, gap, and five-request time are reported by session; no rerun raw log is identified in the source. First-run objective, bound, gap, and time are transcribed from `pilot/work/v2_tests/v2_results.txt`. The 3,600 s rerun limits for 10 and 15 requests follow the documented one-hour limit; a precise termination time is not given. SA was reported as 4.6% above v2's proven five-request optimum and at least 20% above the unproven ten-request v2 solution. A full SA/v2 objective-and-vehicle table is **[OPEN-A5]**.

4.3 Generation profile

The six-request profile took 250 seconds wall time and 192 seconds CPU before the cache. Nearest-node lookup accumulated 117.8 seconds of the 154.3-second request phase; imports took about 60 seconds and the request plot about 27 seconds. After the fix, the whole run took 115 seconds wall time and 72 seconds CPU. The reported post-fix maximum memory of 67 MB is suspicious and must not be used for sizing [**OPEN-G6**].

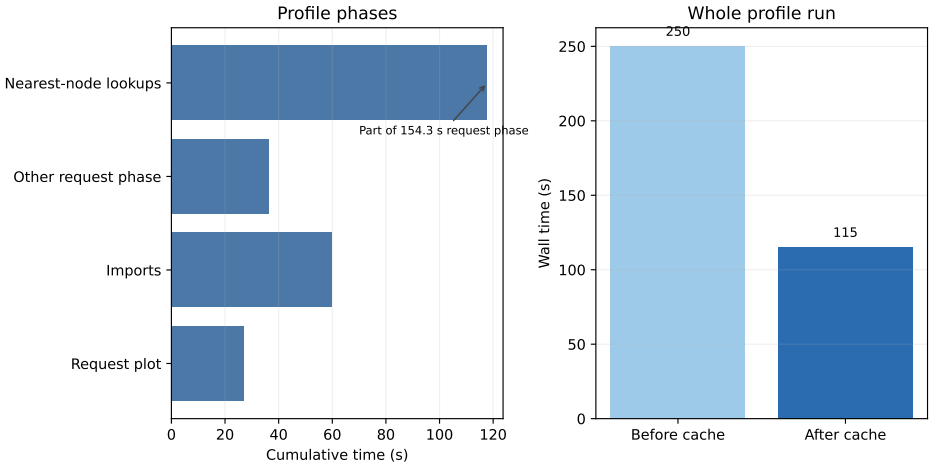


Figure 3: Measured generation costs before the nearest-node cache and total wall time before/after it. The nearest-node bar is part of the full request phase, not an additional phase. After-fix total uses the documented 115 s measurement. Source: `docs/02_tests_and_results.md`, T7.

4.4 Network build and resource use

The region network contains 3515 stations, with none on Bornholm. The build used one core and ran for 3.4 hours while waiting on Overpass for part of the time. A separate generation job with a station matrix had used 7.1 GB of an 8 GB reservation after 5.3 hours and had written no output files. The eventual job result is [**OPEN-G1**]; the unaccounted time is [**OPEN-G3**].

Table 5: Recorded HPC resource use. Reserved memory is per slot. Missing or unmeasured values are marked rather than estimated.

Phase	Cores	Reserved	Used	CPU use	Runtime
Network build	1	6 GB	4466 MB	28%	12374 s (3.4 h)
6-request profile, before fix	1	8 GB	1593 MB	77%	250 s
6-request profile, after fix	1	8 GB	67 MB[a]	63%	115 s
218-request generation test	1	8 GB	7.1 GB	About 1 core	5.3 h, unfinished
SA/GR pilot solver	1	16 GB	About 6 GB	Not reported	Not reported
Exact model v2	4 threads[b]	Not stated	Not stated	Not stated	1 h limit

[a]The 67 MB maximum is identified as suspicious in the source. [b]The first-run raw logs show four threads; the HPC resource table says the exact-model resource size is not yet established. The exact-model rerun status and timing are reported by session. For network and profile measurements, values are transcribed from the documented measurements; the raw session logs are not all identified. Do not treat unreported solver CPU use or runtime as zero.

5 Decisions log

Table 6: Short form of the decisions recorded in `docs/01_decisions.md`.

ID	Decision
D1	Optimise passenger ride time plus a four-times direct-time penalty for third-party service; operator cost is post hoc.
D2	Use SA and GR in the ISA performance matrix; use exact Gurobi only for validation on 5–15 requests.
D3	A third ISA algorithm is wanted, but its exact variant is [**OPEN**].
D4	Use the matheuristic to identify where SA is weak, not as an ISA algorithm.
D5	Use the Capital Region of Denmark without Bornholm.
D6	Use a two-hour period starting 12:00–19:30.
D7	Use 50–500 requests per hour (100–1000 per instance).
D8	Draw continuous settings with Latin hypercubes, not a small factorial grid.
D9	Grow the set in batches until space growth is small or a cap is reached.
D10	Do not vary depots or hotspots during generation; choose the depot later.
D11	Draw three fleets per request instance with a second Latin hypercube.
D12	Use the documented walking, vehicle-speed, and direct-trip-time settings.
D13	Write the shared station travel-time matrix once per network.
D14a	Keep the current request range for now; it may be raised later.
D14b	Compute the station matrix in a separate HPC job.
D14	Keep settings editable in `isa_set/design.json`; final choices belong to the author.
D15	Replace the earlier IP model with model v2.
D16	Make request seeds depend on the config seed.
D17	Restrict station, school, and hospital lookups to the network polygon.
D18	Use a cached nearest-node index in request sampling.
D19	Preserve directed travel times and convert walking distance to seconds.
D20	Keep hybrid-feasibility code but deprecate it for this ISA.
D21	Run solvers and long generation jobs on DTU HPC, not the author's PC.
D22	Reserve only measured resources with modest headroom; report CPU, wall time, and maximum memory.

6 Open questions and next steps

The table consolidates every open item in the source notes. Repeated mentions of the same issue (for example the station-matrix cost in the test results and the open-items list) appear once. Items marked done in the notes are not relisted as open.

Table 7: Open items and whether they block the full set or a later analysis.

ID	Item	Blocking?
OPEN-G1	Determine the final result of test job 29846018; it reached the six-hour limit with no output files.	Yes: full generation
OPEN-G3	Measure where the test job's unaccounted ∼4 hours went; run the separate station-matrix job and record time and memory.	Yes: full generation

ID	Item	Blocking?
OPEN-G4	Implement the feature step that attaches the shared station matrix, required for consistent time-window features.	Yes: feature pipeline
OPEN-G5	Choose batch memory and wall-time settings from a measured full instance.	Yes: batch runs
OPEN-G6	Resolve the suspicious 67 MB maximum-memory report after the nearest-node fix; do not size from it.	No: verify before sizing
OPEN-A1	Choose the third ISA algorithm (SA configuration or max-ALNS) and identify its code.	No: ISA algorithm set
OPEN-A2	Run the MIP-based matheuristic on HPC and record results.	No: matheuristic study
OPEN-A3	Resize SA/GR memory reservation (16 GB reserved; about 6 GB used).	No: solver efficiency
OPEN-A4	Commit and rebuild the solver with `maxstations` 4000.	Yes: Capital Region solver run
OPEN-A5	Tabulate SA and v2 objectives, bounds, and vehicles used on the 5- and 10-request cases.	No: validation report
OPEN-A6	Test whether SA's 70-of-72 advantage over GR persists on the wide design.	No: full-set analysis
OPEN-S1	Revisit the request range if generation cost allows going above 50–500 per hour; the current range is decided for now.	No: later design revision
OPEN-S2	Decide whether maximum walking time should vary, given its effect on candidate stops.	No: design choice
OPEN-S3	Decide whether several depots affect algorithm-performance differences.	No: design choice
OPEN-S4	Set the cap for batches of 100 instances.	Yes: stopping rule
OPEN-D1	Add a description of what `REQreate/rank_model.py` computes.	No: documentation
OPEN-D2	Diagnose or document the seven web-app tests that fail on this branch and its predecessor.	No: unrelated test suite
OPEN-D3	Determine why the first generation job-array attempt passed an empty config argument.	No: replaced workflow
OPEN-D4	Clarify that the `/jobs` reminder tracks only assistant-run `bsub` commands.	No: monitoring
OPEN-T3	Establish resource sizing for the exact-model jobs; the source lists threads and memory as not yet sized.	No: exact-model scale-up

The same next-step priorities follow from the evidence: complete and measure the station-matrix path before scaling generation; resolve memory reports before setting reservations; then expand the instance set and test algorithm selection on the continuous design. The author still makes final choices for the [**PROPOSED**]settings.

7 Reproduction

The full generation and solver pipeline is intended for DTU HPC. These commands are transcribed from `docs/05_hpc_and_reproduction.md`; they are not to be run as local solver or generation jobs.

```
# Build the network once
```

```
bash ~/instance-generator/isa-features-plan-implementation/isa_set/hpc/build_network.
    sh

# Draw a batch of 100, then submit one single-core job per config
python ~/instance-generator/isa-features-plan-implementation/isa_set/make_batch.py --
    batch 1 --n 100
bash ~/instance-generator/isa-features-plan-implementation/isa_set/hpc/generate_batch.
    sh 1

# Profile a small generation run
bsub -J isa_prof -q hpc -n 1 -R "rusage[mem=8GB]" -M 8GB -W 1:00 \
  -o /work3/$USER/isa/logs/profile.out \
  bash ~/instance-generator/isa-features-plan-implementation/isa_set/hpc/profile_one.
    sh <config.json> 6
```

`MEM_GB` and `WALL_H` override the 8 GB and 6 hour generation defaults. The recorded log pattern is `/work3/<user>/isa/logs/batch_XX/gen_<instance>.out`. The documented pipeline then collects feature rows, runs SA and GR for every instance/fleet, projects the set with `src/isa_pipeline.py`, and measures space growth. The matrix attachment step needed for the time-window features is not written yet [**OPEN-G4**].

8 Source and verification notes

8.1 Figures and their data sources

- **Figure 1, pilot instance space:** `pilot/work/results/instance_space.csv` and `pilot/work/results/perf_matrix.csv`.
- **Figure 2, SA/GR ratio by fleet and time window:** `pilot/work/results/perf_matrix.csv`.
- **Figure 3, generation profile:** measured values in `docs/02_tests_and_results.md`, T7. The after-fix total wall-time bar is from the measured 115-second rerun. There is no source CSV for the profile timings.

All three PDFs are regenerated by `make_figures.py`. The seed-count comparison is intentionally text-only because the available pre-fix data do not reproduce the documented value.

8.2 Numbers not independently verified from raw records

- The pre-fix count of 6 distinct `od_distance_mean` values is reported in the docs but not reproduced by the available data. The available pre-fix-labelled CSV has 24 rows and 3 distinct values; the post-fix files have 24 distinct values.
- The rerun exact-model result (5 requests optimal in 2.5 s, 10 requests at 25% gap, and 15 requests not optimal) is reported by session. The source does not identify a raw rerun log, nor tabulate rerun objectives or bounds.
- The 67 MB post-fix maximum-memory result is expressly described as suspicious in the source and is not a reliable memory estimate.
- The final outcome of generation job 29846018, the measured station-matrix time and memory, the cause of the extra generation time, and resource sizes for subsequent batches are not available.
- The exact-model objectives and bounds for the rerun, and the comparative SA/v2 vehicle-use table, remain unreported.

8.3 Source inconsistencies to resolve

- The feature list gives general request-count ranges of 5000–20000 and fleet-size ranges of 15–25%; the current study design instead uses 100–1000 requests per instance and 5–40% fleet size. The study-specific design is taken from `isa_set/design.json`, but the older feature range should be reconciled in the source notes.
- The paper-settings file describes the pilot period as 6:30–8:30, while the generator-change notes describe pilot v2 as having a real 7:30 peak and the current design as low-demand starts between 12:00 and 19:30. These may refer to different protocol versions; the source set does not fully explain the difference.
- The docs report six pre-fix distinct mean OD distances over 72 pilot cases, but available `pilot/work/pilot_v1/results/features.csv` contains 24 rows and 3 distinct values. The 72-row files labelled `pilot_v2_old` and `results` each contain 24 distinct values. The files therefore do not reproduce the stated before-fix/after-fix comparison as described.
- The design JSON names the network as the Capital Region but does not encode the Bornholm exclusion in the network string. The decisions and network test specify the exclusion, so it is treated as part of the design here.

References

[1] Queiroz and Sorensen. Evaluating a simulated annealing metaheuristic for the heterogeneous on-demand bus routing problem with a novel experimental setup. R1, 2026.

[2] Queiroz and Sorensen. Parallel simulated annealing metaheuristic for the heterogeneous on-demand bus routing problem. *Parallel Computing*, 129:103212, 2026.